

Lines & Angles

Use the cue, then fade the cue. Study the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

CUE FOCUS A **point** is an exact spot. A **line** goes on forever in both directions. A **ray** starts at one point and goes on forever in one direction. A **line segment** has two endpoints. **Parallel** lines stay the same distance apart and never cross; **perpendicular** lines cross to make a square corner. Angles: a **right angle** is a square corner (like the corner of a page), an **acute angle** is smaller than a right angle, and an **obtuse angle** is larger than a right angle.

Round 1 — Retrieve It

1. What is a line segment?

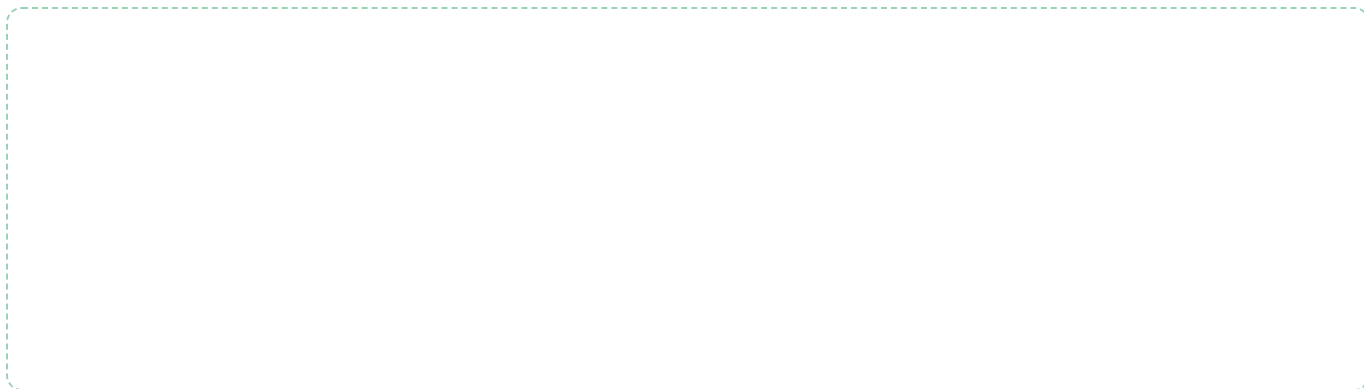
2. What do we call two lines that cross to make a square corner?

3. Which angle is smaller than a right angle?

4. How is a ray different from a line?

Round 2 — Sketch It

Draw three boxes labeled right angle, acute angle, and obtuse angle. Draw and label each type of angle inside its box.



Lines & Angles — Apply It

Keep the cue card covered. Answer first. Then use the cue to check, revise, and explain what you changed.

Round 3 — Apply the Idea

1. The two long sides of a ruler never meet and stay the same distance apart. What kind of lines are they?

2. The corner of a book makes a square corner. What kind of angle is that?

3. An angle opens wider than a square corner. Is it acute or obtuse? How do you know?

Round 4 — Reason from Evidence

A friend drew two lines that cross but not at a square corner and called them perpendicular. Explain what perpendicular really means and why the friend is wrong.

Lines & Angles — ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A**Articulate**

Explain the difference between a line, a ray, and a line segment.

C**Connect**

Connect parallel or perpendicular lines to something you have seen, such as train tracks or a window frame. Tell which kind it shows.

E**Extend**

Find a right, an acute, and an obtuse angle in your classroom. Name each object and tell which angle it makes.

Confidence check. Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it.

Teacher Key — Lines & Angles

Remove this page before student copying. Answers below give expected ideas, not required wording.

Retrieve It

1. A straight path with two endpoints.
2. Perpendicular lines.
3. An acute angle.
4. A ray starts at one point and goes on forever in only one direction; a line goes on forever in both directions.

Sketch It — look for

Right angle drawn as a square corner, acute angle drawn smaller (more closed), and obtuse angle drawn wider (more open) than the right angle.

Apply It

1. Parallel lines.
2. A right angle.
3. Obtuse; it opens wider (larger) than a right angle.

Reason from Evidence — look for

Perpendicular lines must cross to form a square corner (a right angle). Lines that cross at a slanted angle are just intersecting lines, not perpendicular.

ACE — Extend look-for

Right: corner of a desk, book, or window. Acute: an open pair of scissors slightly open, or the tip of a slice. Obtuse: an open laptop or a wide-open door angle.

Suggested use. Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color.